

ABIDE BY ROHS

■ Specification :

●1.Rating:

Voltage :12V AC/DC max.,2V DC min.

Current: 10mA AC/DC max.,10μA DC min.

●2.Contact Resistance : 100mΩ Max

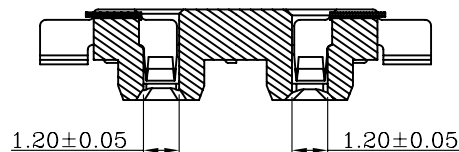
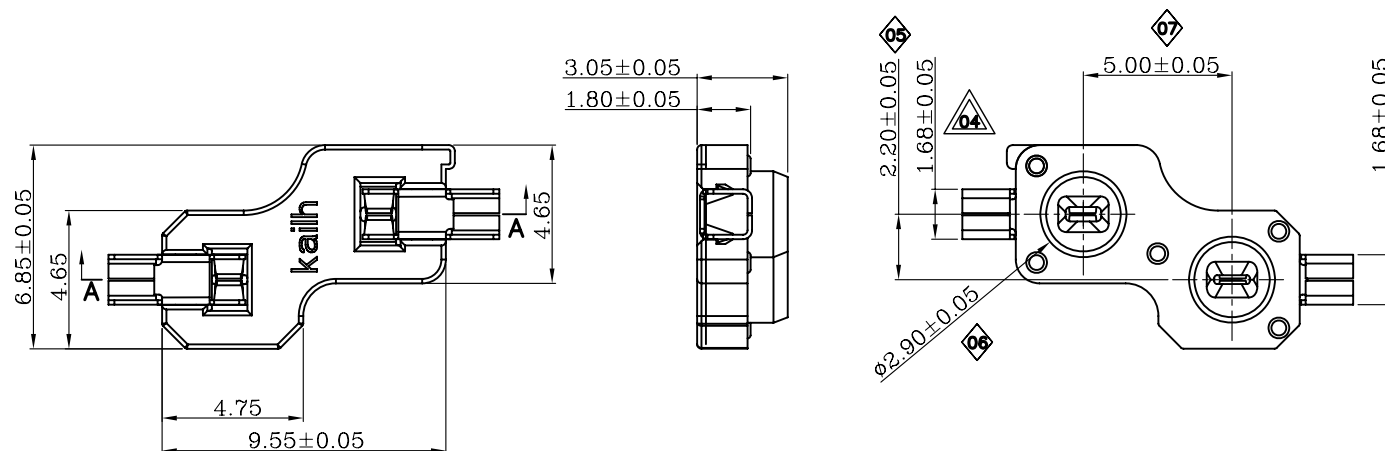
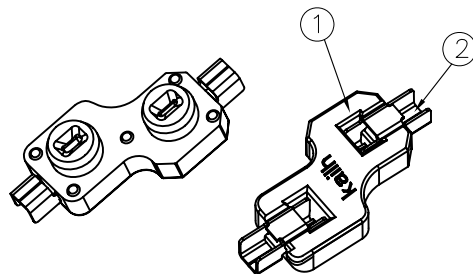
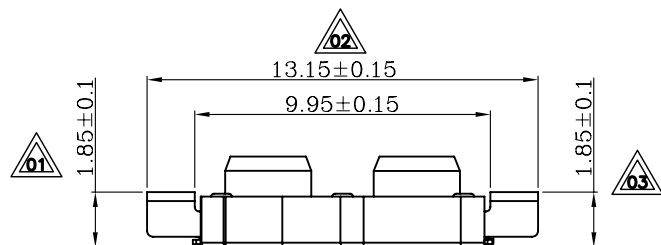
●3.Insulation Resistance :100MΩ min (500V)

●4.Withstand Voltage : AC100V(50-60Hz)  
for 1 minute

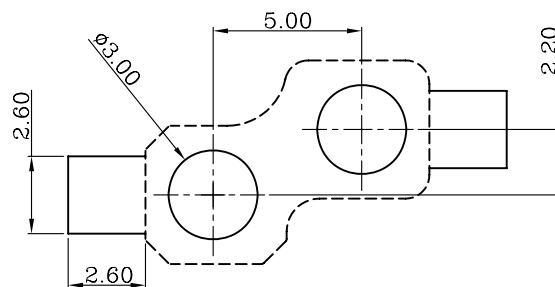
●5.Contact:All the Surface plating Ni,then gold  
plating for contact Area, Tin plating for welding  
area

●6.Mating Force :3.0kgf max

●7.Mechanical Life :100 Cycles



SECTION A-A



PCB LAYOUT

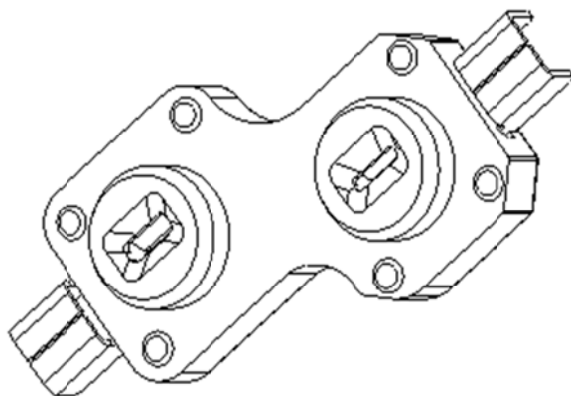
|                |           |        |            |                                          |             |                            |
|----------------|-----------|--------|------------|------------------------------------------|-------------|----------------------------|
| ②              | contact   | —      | 2          | Copper Alloy                             | —           | —                          |
| ①              | Base      | —      | 1          | Nylon                                    | Black       | —                          |
| ITEM           | PART NAME | TER'NO | QTY.       | MATERIAL                                 | FINISHING   | REMARK                     |
| APPROVALS      |           |        | DATE       | DONGGUAN CITY KAIHUA ELECTRONICS CO.,LTD |             |                            |
| DRAWN          |           |        | huyuanfeng | 2017-01-26                               | Kaih        |                            |
| CHECKED        |           |        |            |                                          | TITLE:      | PG1350 KeySwitches Contact |
| APPROVALS      |           |        |            |                                          | PART NO.    | CPG135001S30               |
| TOLERANCES ARE |           |        | 30<L       | ±0.30                                    | ANGLE       | UNIT: mm                   |
|                |           |        | 10<L≤30    | ±0.20                                    |             |                            |
|                |           |        | 5<L≤10     | ±0.15                                    | ±2°         | SCALE: 1:1                 |
|                |           |        | L≤5        | ±0.10                                    |             |                            |
|                |           |        |            |                                          | DRAWING NO. | KHA-PG1350-005EN           |
|                |           |        |            |                                          |             | SHEET 1 OF 1               |

|         |      |       |              |         |        |        |
|---------|------|-------|--------------|---------|--------|--------|
| —       | A    |       | NEW          | —       | —      | —      |
| ECN NO. | REV. | DATE. | DESCRIPTION. | CHANGE. | CHECK. | APPRO. |



KH-PS-1702-35

# Product Specification



CPG135001S30

## 1350 Connector

[illegible]



凱華電子  
KAIHUA EEELECTRONICS

## Product Specification

|              |               |       |       |
|--------------|---------------|-------|-------|
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凱華電子  
KAIHUA EEELETRONICS

## Product Specification

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### 1. Scope/范围:

This Product Specification covers the requirement of Mechanical keyboard Connector switch on product performance, test methods and quality assurance provisions.

本规格书内容涵盖机械键盘连接器产品的要求，包括性能指标、测试方法及质量保证方面等。

### 2. Product Application/产品应用:

Mainly applied on computer keyboards, cash registers equipment and Man-Machine interface.

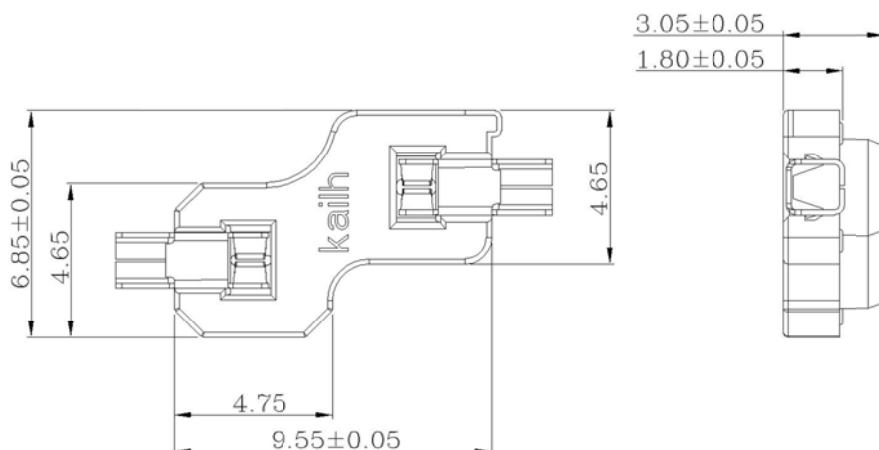
主要适用于电脑，收银机，工业设备和人机界面

### 3. Technology Parameters/技术参数

|                                       |                 |
|---------------------------------------|-----------------|
| Ambient Humidity 工作湿度:                | 45~95% R.H.;    |
| Operating Temperature Range 使用温度范围:   | -10℃~+60℃;      |
| Storage Temperature Range 保存温度范围:     | -20℃~+70℃;      |
| Normal Condition:                     |                 |
| Ambient temperature 环境温度:             | 20±2℃           |
| Relative humidity 相对湿度:               | 85%±5% R.H.;    |
| Air pressure 气压:                      | 86~101KPa;      |
| Contact Resistance 接触阻抗:              | 100 mΩ Max;     |
| Solder Ability 可焊性:                   | 260±5℃, 3±0.5s; |
| Withstand Soldering Temperature 耐焊接热: | 260±5℃, 5±1s;   |

### 4. Ratings/额定性能要求

|                            |                             |
|----------------------------|-----------------------------|
| Rating Voltare 额定电压:       | 12V AC/DC max; 2V DC min    |
| Rating Current 额定电流        | 10mA AC/DC max; 10uA DC min |
| InsulationResistance 绝缘电阻: | ≥100MΩ/DC 500V;             |
| Withstand Voltage 耐电压:     | AC 100V 1 Minute;           |
| Mechanical Life 机械寿命:      | 100Cycles .                 |
| Profile Dimensions /外形尺寸   |                             |



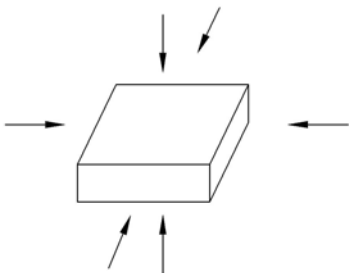


5. Electrical Performance/电气性能

| Item<br>项目 | Description<br>项目描述                          | Test Condition 测试条件                                                                                                                                                                                                  | Requirement 规格要求                          |
|------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 6.1        | Contact<br>Resistance<br>接触电阻                | Static load: (Operation force)x2, which is applied on the center of Switch stem.<br>静态负载: 动作力的 2 倍, 施加在手柄中心.<br><br>Measurement tool: Contact resistance Meter.<br>测量工具: 微电流接触电阻计(1KHz, 20mV,5~50mA)                 | 100mΩ Max<br>100mΩ 以下                     |
| 6.2        | Insulation<br>Resistance<br>绝缘电阻             | Apply a Voltage of DC 500 V for 1 minute, according to the below method.<br>(1) Between terminals.<br>(2) Between terminal and Body.<br><br>输入 500V DC 电压 1 分钟, 按如下接触方法测试:<br>(1) 端子与端子之间.<br>(2) 端子与外壳之间.           | 100MΩ Min<br>100 兆欧以上                     |
| 6.3        | Dielectric<br>withstanding<br>voltage<br>耐电压 | Apply a Voltage of AC 100 V (50~60Hz) for 1 minute, according to the below method.<br>(1) Between terminals.<br>(2) Between terminal and Body.<br><br>输入 100V AC 电压 1 分钟, 按如下接触方法测试:<br>(1) 端子与端子之间.<br>(2) 端子与外壳之间. | No evidence of<br>breakdown<br>无瞬断、击穿等破坏. |



6. Mechanical Performance/机械性能

| Item<br>项目 | Description<br>项目描述   | Test Condition 测试条件                                                                                                                                                                                                                                                                     | Requirement 规格要求                                                            |
|------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 7.1        | Mating force<br>插入力   | At 16in/minute.actuation speed                                                                                                                                                                                                                                                          | Mating force<br>插入力 3KG max                                                 |
| 7.2        | Unmating force<br>拔出力 | At 16in/minute.actuation speed                                                                                                                                                                                                                                                          | Unmating force<br>拔出力 200gf min                                             |
| 7.3        | Shock<br>机械冲击         | <p>Measured by according to the below condition:<br/>(1) Acceleration: 80g 加速度<br/>(2) Cycles of test:3 cycles each in 6 directions, for a total of 18 cycles.<br/>试验次数: 每个方向 3 次, 6 个方向共 18 次.</p>  | Appearance:<br>No abnormality.<br>外观无异常                                     |
| 7.4        | Life Test<br>寿命测试     | <p>(1) without load 无负载<br/>(2) Mating force: Maximum value of operation force.<br/>插入力: 操作力规格值的上限.<br/>(3) Cycles: 100 Min<br/>操作次数: 100 次以上</p>                                                                                                                                       | Contact resistance:<br>1000 mΩ Max<br>接触电阻: 1000 毫欧以下<br>Bouncing: 10ms Max |



7. Environmental Performance/环境性能

| Item<br>项目      | Description<br>项目描述            | Test Condition 测试条件                                                                                                                                                                                                                                                                                                                                                             | Requirement 规格要求                                                                               |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |
|-----------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------|--------------------------|-----------------|-------|----|--------|----|-------|----|-------|----|------------------------------------------------------------------------------------------------|
| 8.1             | Cold test<br>耐寒性               | (1) Temperature : - 20±2℃<br>温度： - 20±2℃<br>(2) Duration of test: 48h<br>持续时间： 48 小时<br>(3) Take off a drop water 去掉水珠<br>(4) Standard conditions after test : 1h<br>试验后的放置条件： 1 小时                                                                                                                                                                                             | Contact resistance:<br>200mΩ Max<br>接触电阻 200mΩ 以下                                              |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |
| 8.2             | Heat test<br>耐热性               | (1) Temperature : 70±2℃<br>温度： 70±2℃<br>(2) Duration of test: 48h<br>持续时间： 48 小时<br>(3) Take off a drop water 去掉水珠<br>(4) Standard conditions after test : 1h<br>试验后的放置条件： 1 小时                                                                                                                                                                                                 | Contact resistance:<br>200mΩ Max<br>Shall meet :<br>No. 6.2<br>接触电阻 200mΩ 以下<br>满足：<br>No. 6.2 |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |
| 8.3             | Temperature<br>cycle<br>温度循环   | (1) Test cycles:20 cycles<br>试验周期： 5 个周期<br>(2) Standard condition after test:1h<br>试验后的放置条件： 1 小时 <table><tr><td></td><td>Temperature<br/>温度</td><td>Duration of test<br/>持续时间</td></tr><tr><td rowspan="4">1 cycle<br/>一次循环</td><td>20±5℃</td><td>1h</td></tr><tr><td>-20±5℃</td><td>1h</td></tr><tr><td>20±5℃</td><td>1h</td></tr><tr><td>70±5℃</td><td>1h</td></tr></table> |                                                                                                | Temperature<br>温度 | Duration of test<br>持续时间 | 1 cycle<br>一次循环 | 20±5℃ | 1h | -20±5℃ | 1h | 20±5℃ | 1h | 70±5℃ | 1h | Contact resistance:<br>200mΩ Max<br>Shall meet :<br>No. 6.2<br>接触电阻 200mΩ 以下<br>满足：<br>No. 6.2 |
|                 | Temperature<br>温度              | Duration of test<br>持续时间                                                                                                                                                                                                                                                                                                                                                        |                                                                                                |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |
| 1 cycle<br>一次循环 | 20±5℃                          | 1h                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |
|                 | -20±5℃                         | 1h                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |
|                 | 20±5℃                          | 1h                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |
|                 | 70±5℃                          | 1h                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |
| 8.4             | Soldering<br>heat test<br>耐焊接热 | Soldering area: T/2 of PWB thickness.<br>(PWB: T=1.6mm)<br>焊接面积： 印刷基板的 1/2 厚度处<br>Soldering temperature: 260±5℃<br>Soldering time: 5±0.5s<br>焊接温度： 260±5℃<br>焊接时间： 5±0.5 秒                                                                                                                                                                                                      | Appearance:<br>No abnormality.<br>外观无异常                                                        |                   |                          |                 |       |    |        |    |       |    |       |    |                                                                                                |

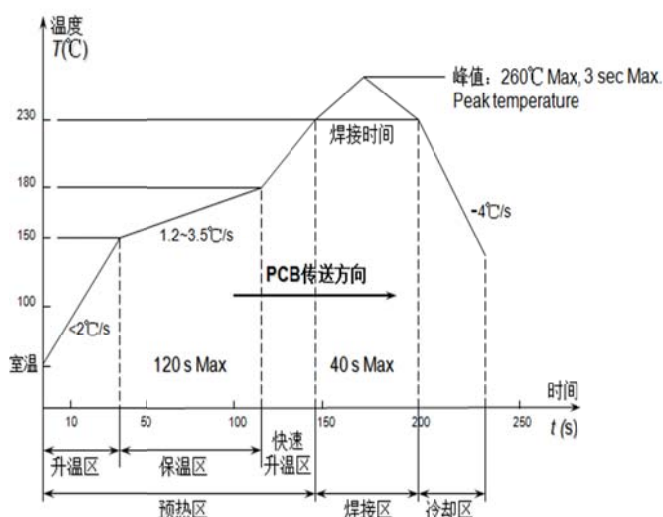


8.5

Solderability  
可焊性

1. Hand soldering 手工焊接:  
Please practice according to below condition:  
(1) Soldering Temperature :  $350 \pm 5^{\circ}\text{C}$   
焊接温度:  $350 \pm 5^{\circ}\text{C}$   
(2) Continual soldering time:  $3 \pm 0.5\text{s}$   
连续焊接时间:  $3 \pm 0.5$  秒  
(3) Capacity of soldering iron:  $\leq 20\text{w}$   
电烙铁功率: 20 瓦以下
2. Automatic Reflow soldering 自动回流焊接:  
For the product of SMT, according to below condition:

SMT 回流焊温度曲线图



At least 95% of surface area of immersed portion shall be covered by solder.  
侵焊面积大于 95% 以上.

8.6

Humidity test  
耐湿性

- (1) Temperature :  $60 \pm 2^{\circ}\text{C}$   
温度:  $60 \pm 2^{\circ}\text{C}$
- (2) relative humidity: 90~95% R.H.  
相对湿度: 90~95% R.H.
- (3) Duration of test: 48h  
持续时间: 48 小时
- (4) Take off a drop water  
去掉水珠
- (5) Standard conditions after test: 1h  
试验后的放置条件: 1 小

Contact resistance:  
 $200\text{m}\Omega$  Max  
Shall meet :  
No. 6.2  
接触电阻  $200\text{m}\Omega$  以下  
满足:  
No. 6.2



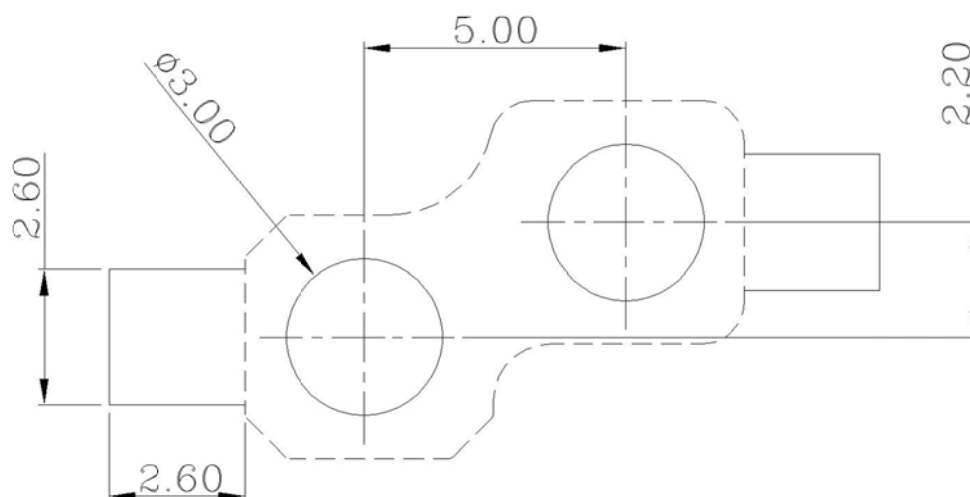
|                                                                                                                      |                              |                            |            |               |
|----------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------|------------|---------------|
|  <b>凱華電子</b><br>KAIHUA EEELETRONICS | <b>Product Specification</b> |                            |            |               |
|                                                                                                                      | P/N:<br>CPG135001S30         | DOC. No.:<br>KH-PS-1702-35 | Rev.:<br>A | Page:<br>8/10 |

|     |                                    |                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                   |
|-----|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.7 | Salt Spray<br>盐雾测试                 | <p>Apply the following environment to test:<br/>根据下列条件进行测试:</p> <p>(1) Temperature : 35±5℃<br/>温度: 35±5℃;</p> <p>(2) Salt water density: 5±1%<br/>盐水浓度: 5±1%;</p> <p>(3) Duration: 12 hours<br/>持续时间: 12 小时;</p> <p>(4) After test, the salt deposit shall be removed by running water.<br/>实验后将盐沉积物用水冲掉</p> | <p>Appearance:<br/>No corrosion spot, no crack, no base plate naked.<br/>外观: 无腐蚀点, 无裂纹, 无裸露基材.</p> <p>Contact Resistance:<br/>200 mΩ Max<br/>接触电阻: 200 毫欧以下</p>   |
| 8.8 | Withstand K <sub>2</sub> S<br>硫化测试 | <p>Apply the following environment to test:<br/>根据下列条件进行测试</p> <p>(1) Temperature: 35±5℃ 温度: 35±5℃</p> <p>(2) K<sub>2</sub>S Density: 2%;<br/>硫化钾浓度: 2%</p> <p>(3) Duration: 2 minute.<br/>持续时间: 2 分钟</p>                                                                                                    | <p>Appearance:<br/>No corrosion spot, no crack, no base plate naked.<br/>外观: 无腐蚀点, 无裂纹, 无裸露基材.</p> <p>Contact Resistance:<br/>1000 mΩ Max<br/>接触电阻: 1000 毫欧以下</p> |

## 8. Recommended PCB Layout 推荐的 PCB 安装焊盘规格

(Top View)

(Single face board T=1.6mm)

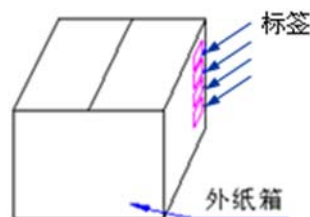
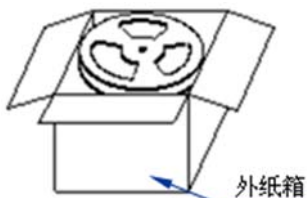
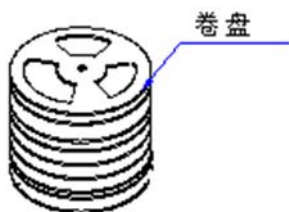




## 10. Packaging 包装

Packaging type: 13Tray, 26000Pcs/Carton.

包装方式:13 /盘, 26000Pcs/箱.



## 11.Precaution 注意事项

### 11.1 Soldering condition 回流焊条件

| ITEM<br>项目                    |                              | CONDITION<br>条件                                                                                           |
|-------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------|
| Preheating zone<br>预热区        | Heating zone<br>升温区          | Speed<2℃/S, Preheating time15 S Max,temperature150℃<br>速度<2℃/S, 预热时间 15 S 最多,温度 150℃                      |
|                               | Heatpreservation area<br>保温区 | Speed1.2~3.5℃/S, Preheating time120 S Max,temperature180℃<br>速度 1.2~3.5℃/S, 预热时间 120 S 最多,温度 180℃         |
|                               | Fast heating zone<br>快速升温区   | Speed3.5~4.5℃/S, Preheating time140 S Max,temperature230℃<br>速度 3.5~4.5℃/S, 预热时间 140 S 最多,温度 230℃         |
| Weld area<br>焊接区              |                              | Welding time 40 S Max, welding temperature peak value, 3 sec Max.<br>焊接时间 40 S 最多, 焊接温度峰值 260℃ 最大, 3 S 最多 |
| Area of flux<br>助焊剂面积         |                              | 1/2 Max of PWB Thickness<br>印刷基板厚度的 1/2 以内                                                                |
| Temperature of solder<br>焊锡温度 |                              | 260±5℃<br>260±5℃                                                                                          |
| Number of soldering<br>焊接次数   |                              | 2time Max (But should down heat of the first soldering)<br>2 次以内                                          |
| Printed wiring board<br>印刷基板  |                              | Single side copper-clad laminates<br>单面铜箔                                                                 |

(1) After reflow, be careful not to clean switches with solvent

回流焊后,注意不要用溶剂清洗.

(2) Under the condition of using soldering iron, soldering temperature shall be 350℃ max within 3 sec.

在使用烙铁的情况下,焊锡温度应在350℃以下,焊接时间3秒以内.



## 11.2 Notes 注意点

- (1) Please be cautious not to give excessive static load connector.  
注意不要施加超负荷的压力或晃动连接器.
- (2) Connector be careful not to stack up P. W. B. after switches were soldered.  
连接器焊接以后,印刷基板注意不要叠放.
- (3) Preservation under high temperature and high humidity or corrosive gas should be avoided  
Especially. When you need to preserve for a long period, do not open the carton.  
保管时尤其应注意避开高湿高温和有腐蚀性气体的环境.如需长时间保存,请不要打开包装箱.
- (4) Products meet the ROHS & REACH environmental management substances control standards  
产品满足 ROHS & REACH 环境管理物质管制标准